

Project: MediCore360 - Healthcare & Patient Management Platform

1. Introduction

MediCore360 is a web-based Healthcare & Patient Management Platform designed for hospitals, clinics, and healthcare providers to unify patient records, appointments, billing, and compliance reporting. The platform consolidates patient demographics, medical history, prescriptions, and billing workflows to provide end-to-end visibility, streamlined operations, and adherence to healthcare regulations.

It supports RESTful APIs with Angular/React frontends, and backend implementations in Java Spring Boot or ASP.NET Core. The architecture emphasizes reliability, secure handling of patient data, and compliance with HIPAA/GDPR standards.

Actors / Users

- Doctor: Reviews patient records, diagnoses, and prescriptions.
- Nurse: Manages patient vitals, care plans, and medication administration.
- Receptionist: Handles patient registration, appointments, and billing.
- Patient: Views medical history, prescriptions, and appointments.
- Compliance Officer: Reviews audit logs and generates regulatory reports.
- Admin: Configures departments, users, and system settings.

2. Module Overview

2.1 Identity & Access Management

Authentication, RBAC, audit trails.

2.2 Patient & Record Management

Patient registration, medical history, and record maintenance.

2.3 Appointment & Scheduling

Appointment booking, rescheduling, and cancellations.

2.4 Billing & Payment Management

Invoices, insurance claims, and payment tracking.

2.5 Prescription & Medication Management

Prescription issuance, medication tracking, and pharmacy integration.

2.6 Compliance & Audit Management

Audit logs, regulatory reporting, and compliance dashboards.

2.7 Analytics & Reporting

Operational dashboards, KPIs, and healthcare reports.

2.8 Notifications & Alerts

Real-time alerts for appointments, prescriptions, and compliance deadlines.

3. Architecture Overview

- **Frontend:** Angular or React SPA for dashboards and workflows.
- **Backend:** RESTful services (Spring Boot / ASP.NET Core).
- **Database:** Relational DB (SQL Server/MySQL/PostgreSQL).
- **Integration:** Insurance systems, pharmacy networks, and regulatory portals.
- **Security:** TLS, encryption at rest, RBAC, audit logging.

4. Module Wise Design

4.1 Identity & Access Management Module

Features

- User provisioning and role assignment
- Secure login with RBAC
- Session management and password policies
- Immutable audit logs

Entities

- **User**(UserID, Name, Role, Email, Phone)
- **AuditLog**(AuditID, UserID, Action, Timestamp)

4.2 Patient & Record Management Module

Features

- Patient registration and demographic capture
- Medical history recording and updates
- Record retrieval and archival
- Patient search and retrieval

Entities

- **Patient**(PatientID, Name, DOB, ContactInfo, MedicalHistory)
- **Record**(RecordID, PatientID, Diagnosis, Treatment, Date, Status)

4.3 Appointment & Scheduling Module

Features

- Appointment booking and rescheduling
- Cancellation workflows
- Doctor availability management

- Calendar integration

Entities

- **Appointment**(AppointmentID, PatientID, DoctorID, Date, Time, Status)

4.4 Billing & Payment Management Module

Features

- Invoice generation
- Insurance claim submission
- Payment tracking
- Refund workflows

Entities

- **Invoice**(InvoiceID, PatientID, Amount, Date, Status)
- **Payment**(PaymentID, InvoiceID, Amount, Date, Method, Status)

4.5 Prescription & Medication Management Module

Features

- Prescription issuance by doctors
- Medication tracking and dosage schedules
- Pharmacy integration
- Prescription renewal workflows

Entities

- **Prescription**(PrescriptionID, PatientID, DoctorID, Medication, Dosage, Date, Status)

4.6 Compliance & Audit Management Module

Features

- Audit log review
- Regulatory reporting generation
- Compliance dashboards
- Exception tracking

Entities

- **ComplianceReport**(ReportID, Scope, Metrics, GeneratedDate)

4.7 Analytics & Reporting Module

Features

- Healthcare dashboards for patient visits and billing
- KPI tracking (Patient Volume, Revenue, Claim Rejections)
- Compliance reporting
- Trend analysis

Entities

- **KPIReport**(ReportID, Scope, Metrics, GeneratedDate)

4.8 Notifications & Alerts Module

Features

- Real-time alerts for appointments and prescriptions
- Escalation workflows
- Configurable notification channels (in-app, email, SMS)
- Notification history

Entities

- **Notification**(NotificationID, UserID, Message, Category, Status, CreatedDate)

5. Deployment Strategy

- **Local / Dev:** Angular/React UI, Spring Boot/.NET backend, local RDBMS.
- **Staging:** Containerized services with synthetic patient and appointment data.
- **Production:** On-premise or hosted deployment with API gateway, managed RDBMS with replicas.
- **High Availability:** Multi-server deployments, automated backups, disaster recovery playbooks.

6. Database Design

Tables

- **User**(UserID, Name, Role, Email, Phone)
- **AuditLog**(AuditID, UserID, Action, Timestamp)
- **Patient**(PatientID, Name, DOB, ContactInfo, MedicalHistory)
- **Record**(RecordID, PatientID, Diagnosis, Treatment, Date, Status)
- **Appointment**(AppointmentID, PatientID, DoctorID, Date, Time, Status)
- **Invoice**(InvoiceID, PatientID, Amount, Date, Status)
- **Payment**(PaymentID, InvoiceID, Amount, Date, Method, Status)

- **Prescription**(PrescriptionID, PatientID, DoctorID, Medication, Dosage, Date, Status)
- **ComplianceReport**(ReportID, Scope, Metrics, GeneratedDate)
- **KPIReport**(ReportID, Scope, Metrics, GeneratedDate)
- **Notification**(NotificationID, UserID, Message, Category, Status, CreatedDate)

7. User Interface Design

- Doctor Dashboard: Patient records, prescriptions, appointments.
- Nurse Console: Patient vitals, care plans, medication schedules.
- Receptionist Dashboard: Patient registration, appointments, billing.
- Compliance Dashboard: Audit logs, regulatory reports.
- Patient Portal: Medical history, prescriptions, appointments.
- Admin Console: Department setup, user roles, system configuration.

8. Non-Functional Requirements

- **Performance:** Handle up to 30,000 concurrent users; appointment booking latency under 200 ms.
- **Security:** RBAC, TLS, encryption at rest, HIPAA compliance, immutable audit logs.
- **Scalability:** Horizontal scaling for patient and appointment services.
- **Availability:** Target 99.9% uptime for healthcare services.
- **Maintainability:** Modular APIs, versioning, automated migrations, CI/CD pipelines.
- **Observability:** Centralized logs, metrics, and alerting.
- **Compliance:** Configurable retention policies, audit trails, regulatory reporting formats.

9. Assumptions & Constraints

- **Phase 1 scope:** Patient registration, record management, appointment scheduling, billing, and dashboards.
- **Excluded in Phase 1:** AI-based diagnostics, telemedicine, advanced mobile health features.
- **Data availability:** Patient and billing data feeds available for integration.
- **Integration:** Insurance systems, pharmacy networks, and regulatory portals via APIs.
- **Implementable:** Using Java Spring Boot or ASP.NET Core, Angular/React, and SQL Server/MySQL/PostgreSQL.